Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.21889365862718857, median 0.2042653253606195, std: 0.12217756873926024
Gyroscope error (imu0): mean 0.06219296201502801, median 0.043548318761689284, std: 0.07214250950862074
Accelerometer error (imu0): mean 0.0689232989769656, median 0.05070300965319785, std: 0.09098646248601

Residuals

Reprojection error (cam0) [px]: mean 0.21889365862718857, median 0.2042653253606195, std: 0.12217756873926024
Gyroscope error (imu0) [rad/s]: mean 0.0024556359890794376, median 0.0017194681737343095, std: 0.0028484853744233565
Accelerometer error (imu0) [m/s^2]: mean 0.026759327825234495, median 0.019685338298873215, std: 0.03532530528080553

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.00354554 -0.99646305 0.08395729 0.01227128]
 [0.04884574 -0.08403017 -0.9952653   0.03729602]
 [0.99880004 0.00057221 0.04897091 -0.04835022]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.00354554 0.04884574 0.99880004 0.04651396]
 [-0.99646305 -0.08403017 0.00057221 0.01538954]
 [0.08395729 -0.9952653   0.04897091 0.03845693]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.00036335094413177983

Gravity vector in target coords: [m/s^2]
[0.36930264 -9.79879001 -0.12551031]

Calibration configuration
==========

cam0

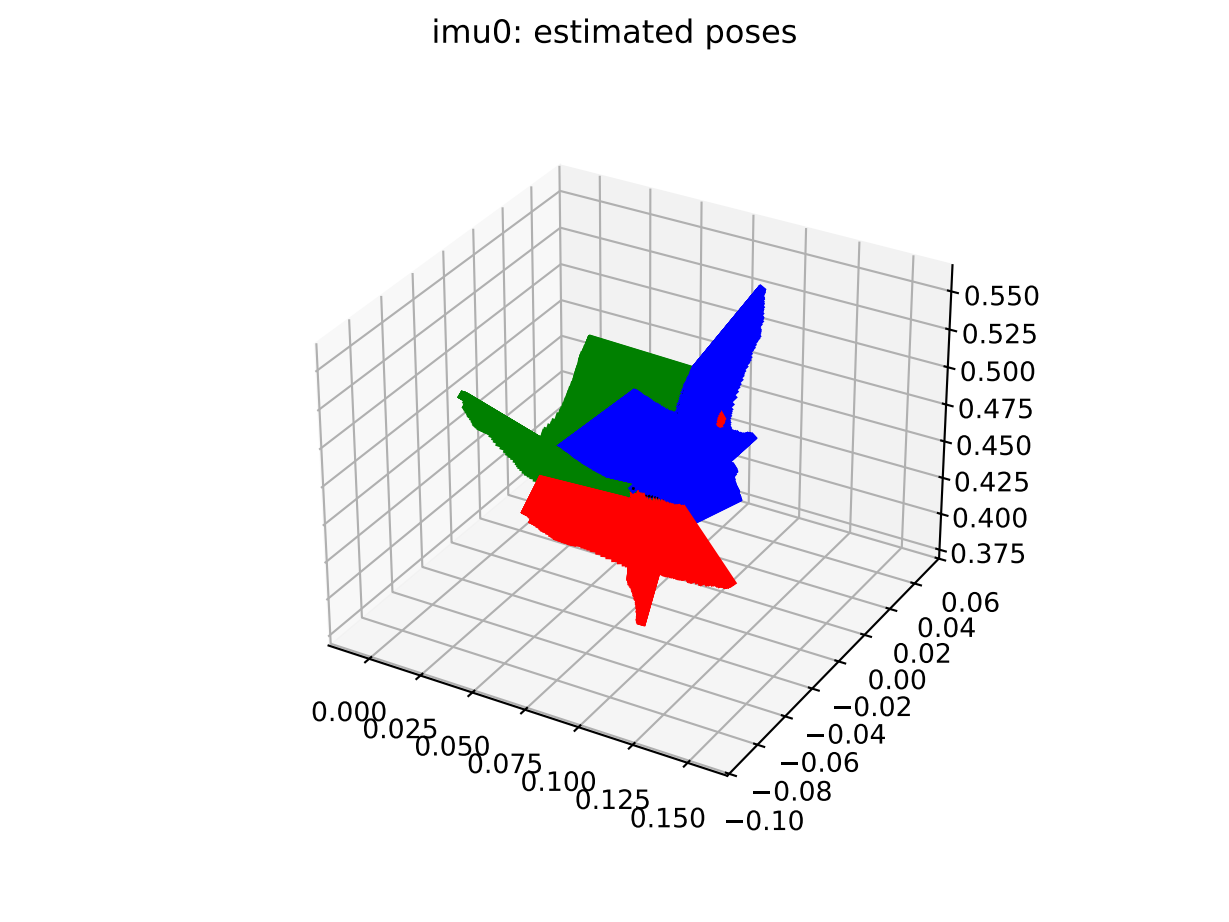
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

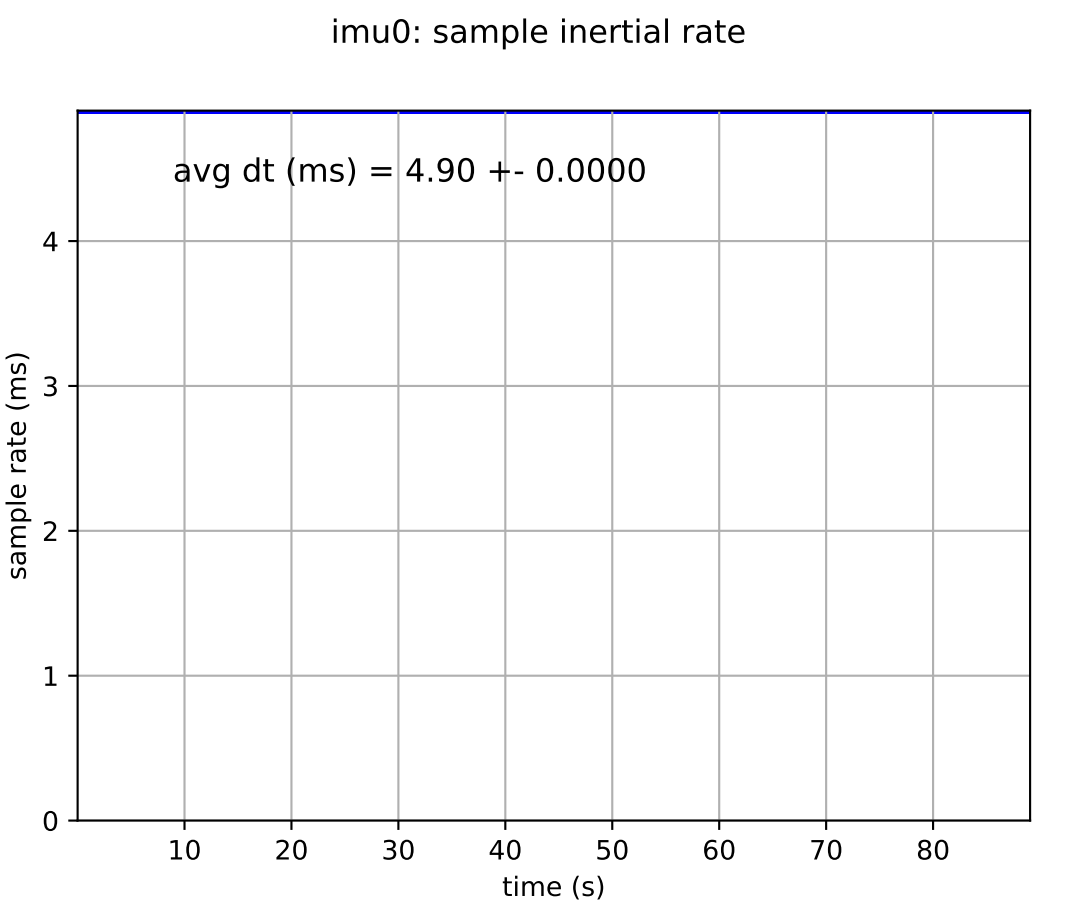
IMU configuration
==========

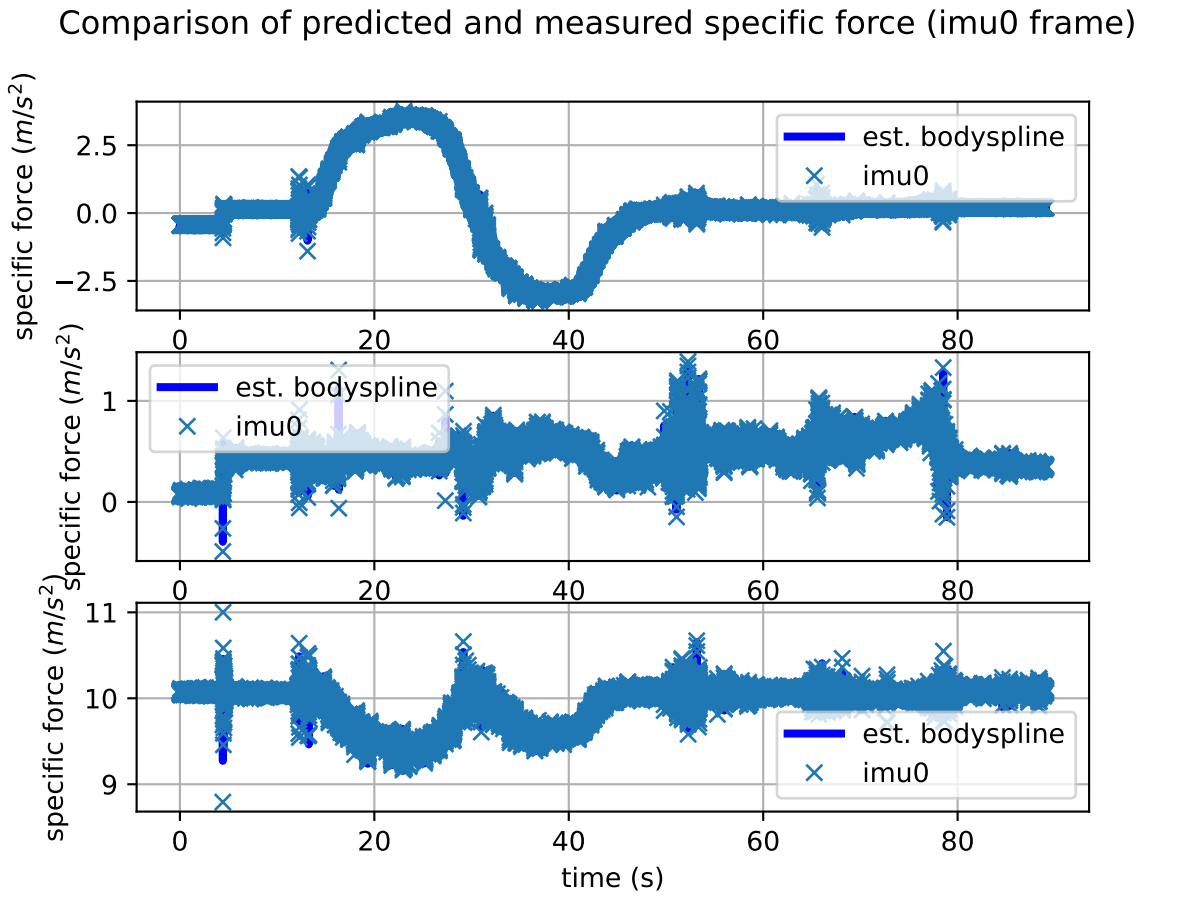
IMU0:

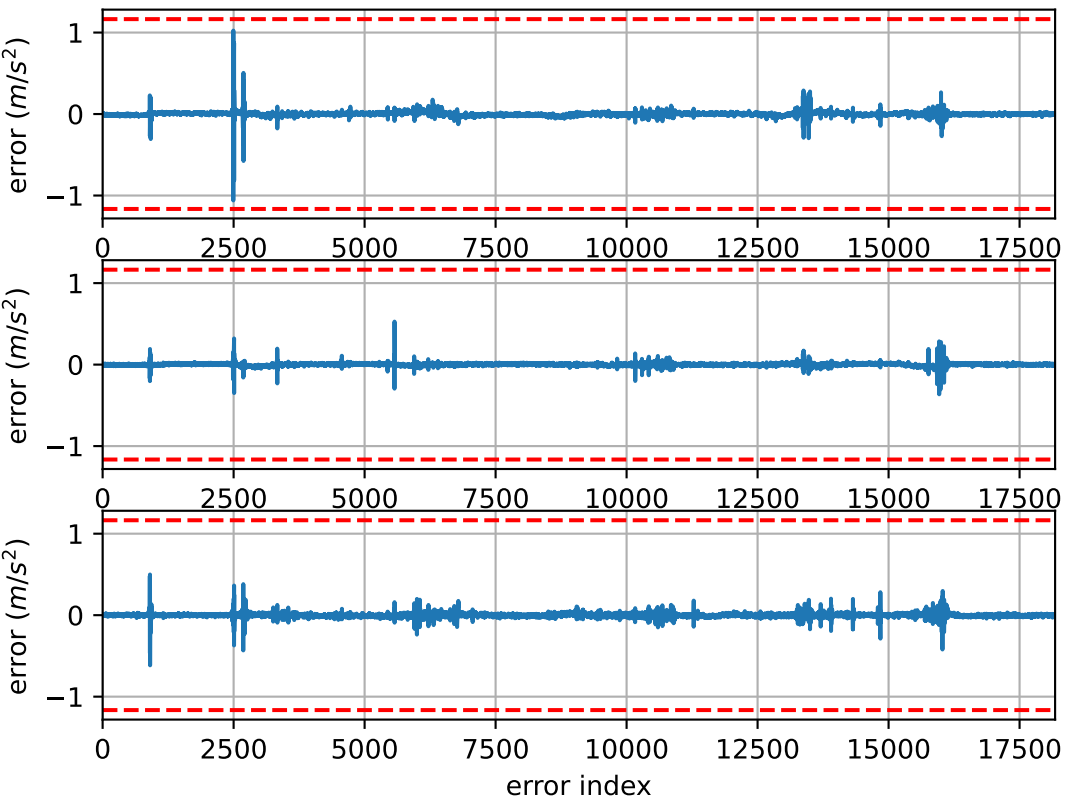
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

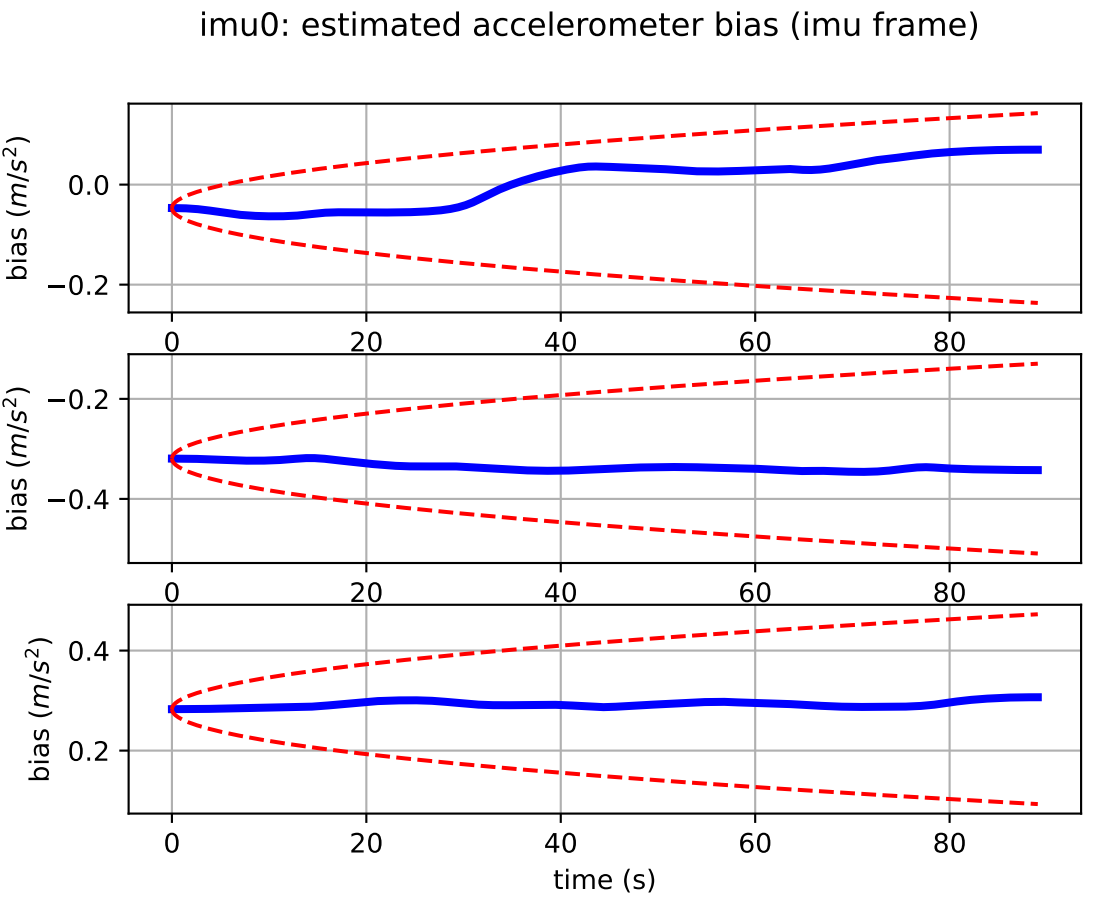
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

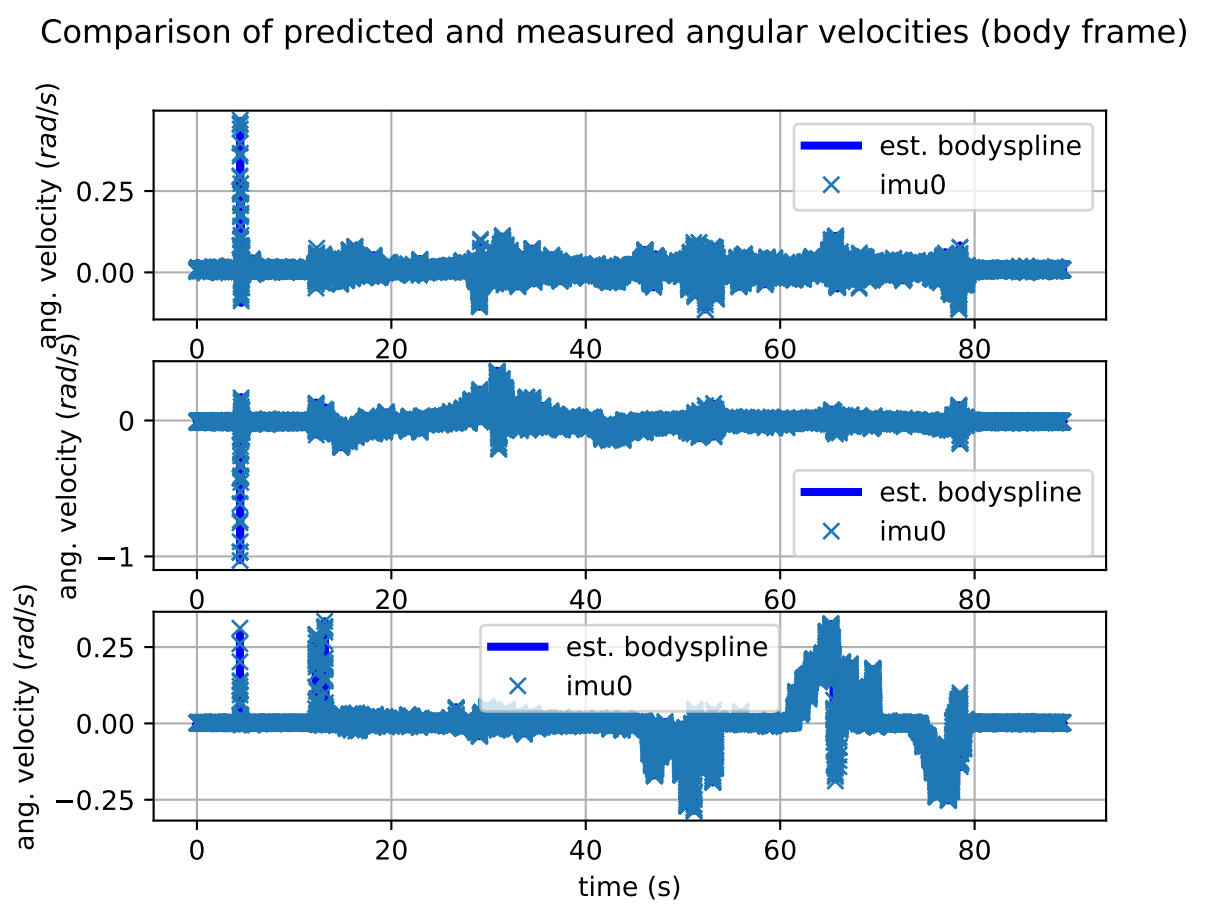

imu0: estimated poses
0.550
0.525
0.500
0.475
0.450
0.425
0.400
0.375
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
−0.10
0.000
0.025
0.050
0.075
0.100
0.125
0.150

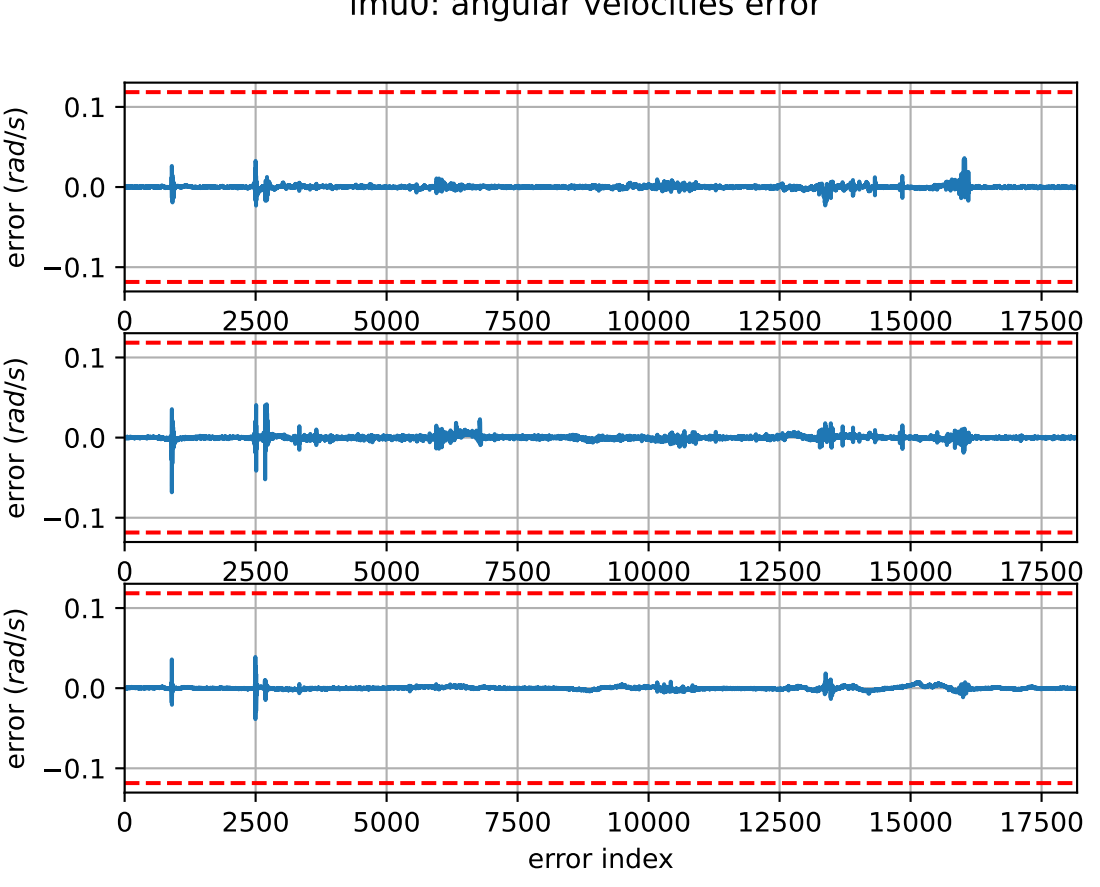

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
60
70
80
time (s)

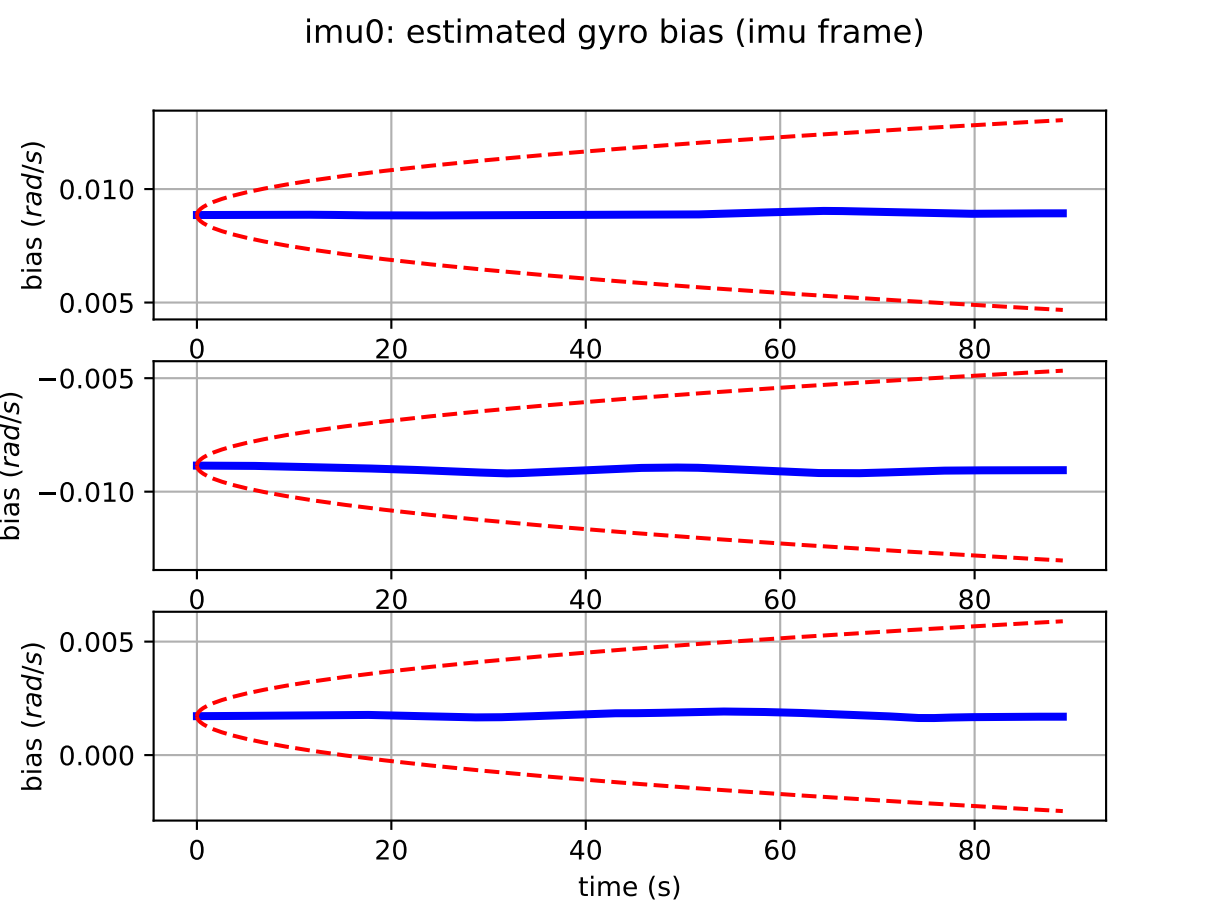

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
2.5
0.0
−2.5
1
0
11
10
9
0
20
40
60
80
time (s)

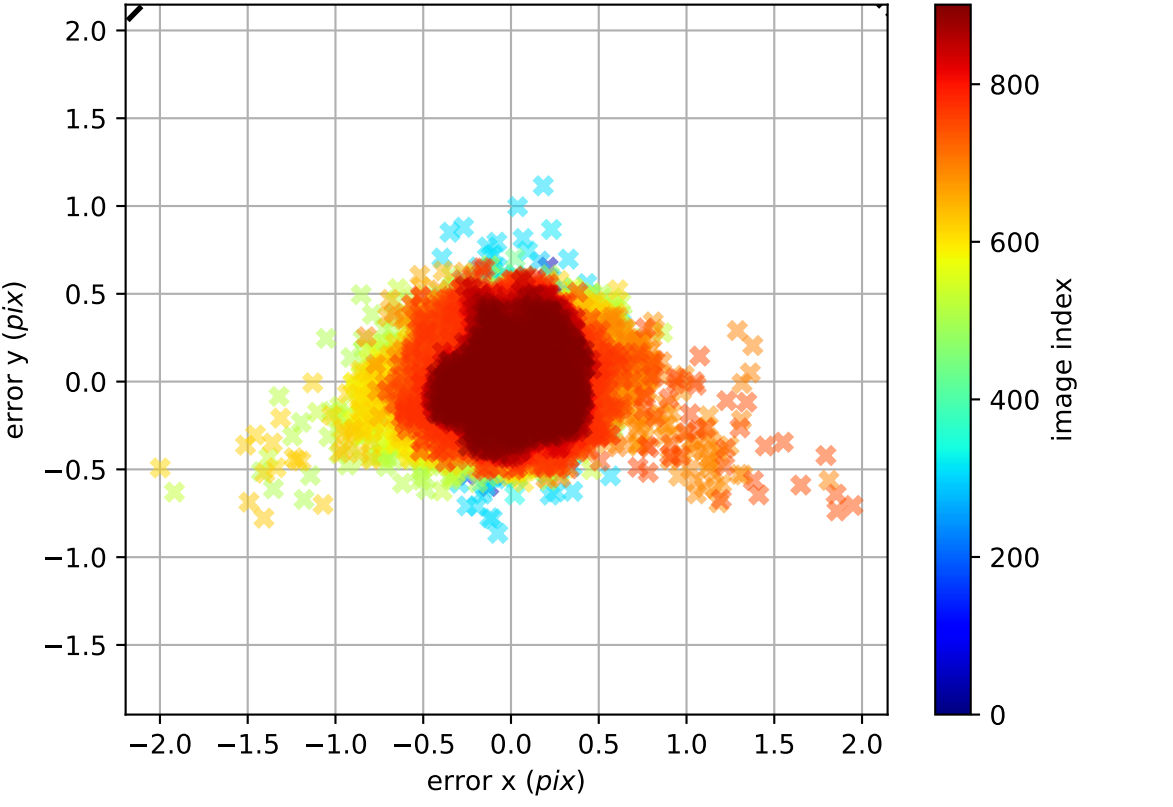

imu0: acceleration error
error (m/s^2)
1
0
−1
0
2500
5000
7500
10000
12500
15000
17500
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.0
−0.2
0
20
40
60
80
bias (m/s^2)
−0.2
−0.4
0
20
40
60
80
bias (m/s^2)
0.4
0.2
0
20
40
60
80
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
0.25
0.00
est. bodyspline
imu0
0
20
40
60
80
ang. velocity (rad/s)
0
−1
est. bodyspline
imu0
0
20
40
60
80
ang. velocity (rad/s)
0.25
0.00
−0.25
est. bodyspline
imu0
0
20
40
60
80
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
2500
5000
7500
10000
12500
15000
17500
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.010
0.005
0
20
40
60
80
bias (rad/s)
−0.005
−0.010
0
20
40
60
80
bias (rad/s)
0.005
0.000
0
20
40
60
80
time (s)


cam0: reprojection errors
error y (pix)
error x (pix)
image index
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
800
600
400
200
0